Université Grenoble Alpes

Regression Modeling Overview

Smart Analytics for Big Data

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Grenoble INP - Genie Industriel

Overview

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Explanatory power

Prediction power

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Introduction

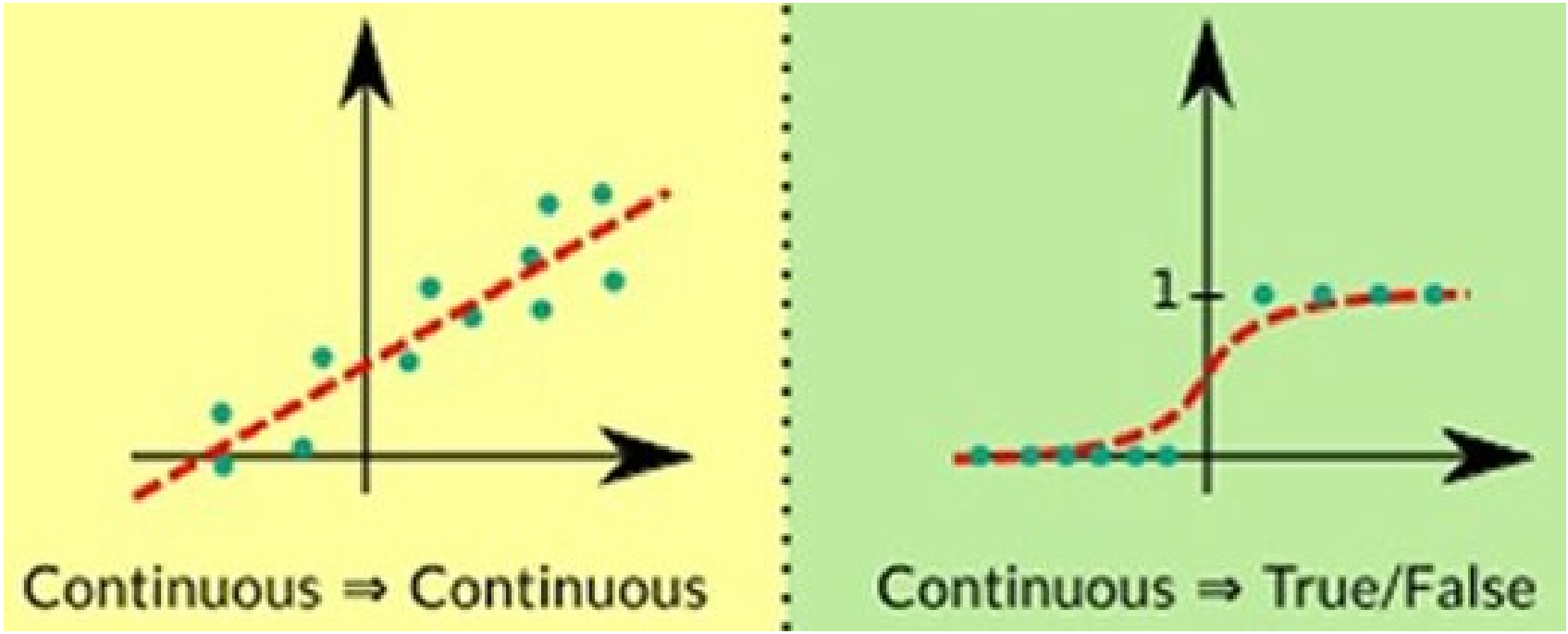
Regression Analysis in application

Objectives

- Overview of regression methods
  - Regression modelling
  - Anatomy of a regression model
- Fitting a model
  - OLS case
- Diagnostic of a model
- Explanatory power
- Prediction power

Regression technics

Regression can be linear, logistic. . .



applied to count data, survival time, time series, panel data,. . .

Regression technics

They share the same objective and issues

- To model the conditional expected outcome ($E(Y|X)$ or $P(Y|X)$)
- using a well chosen stochastic model
  - composed of a functional form: link between X and Y (ex: $Y = \alpha + \beta X + \epsilon$)
  - random disturbance (ex $\epsilon \sim \mathcal{N}(0, 1)$)
  - variables selection and eventually transformation
- to **predict** and to **explain**

Regression technics

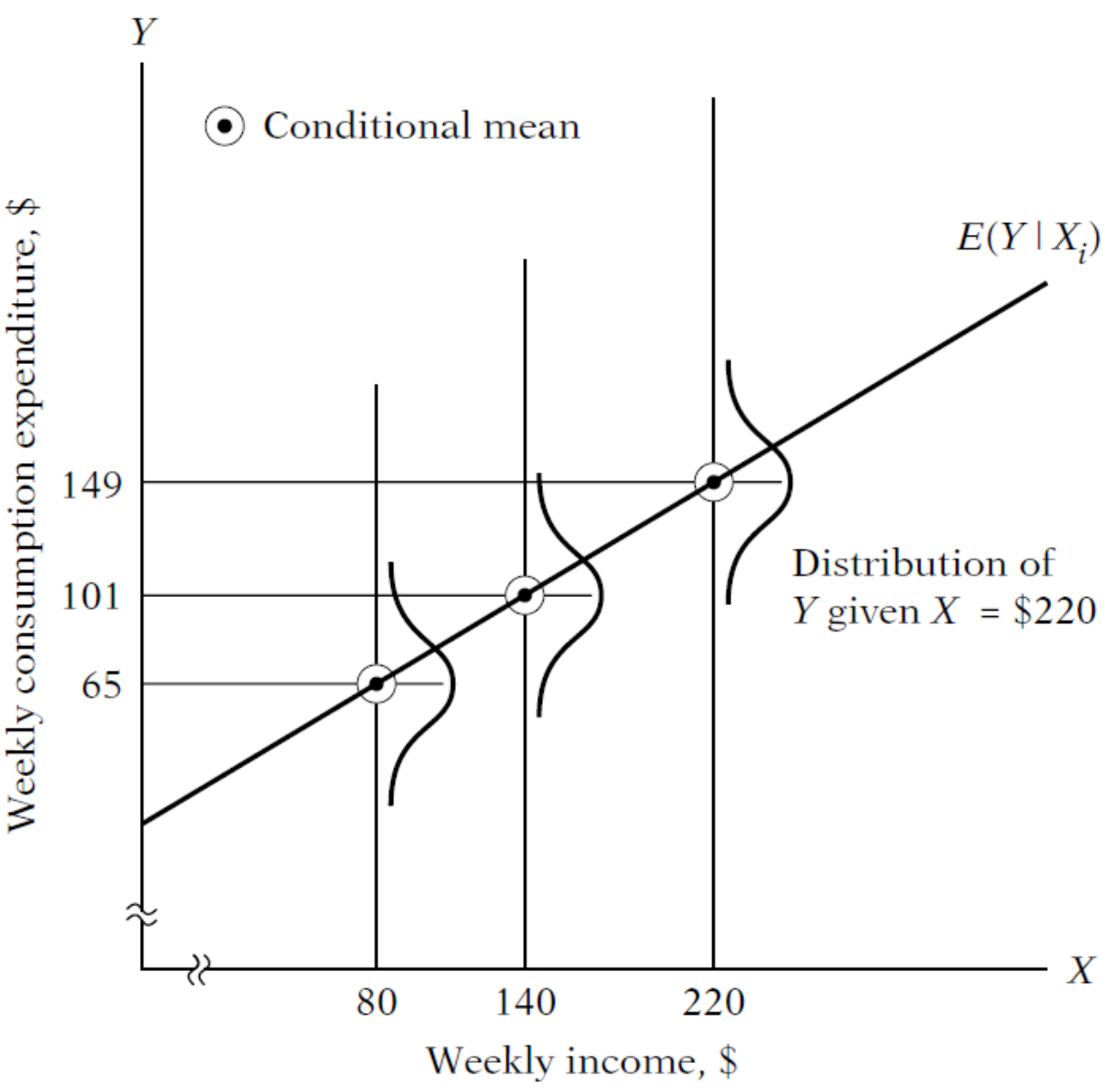
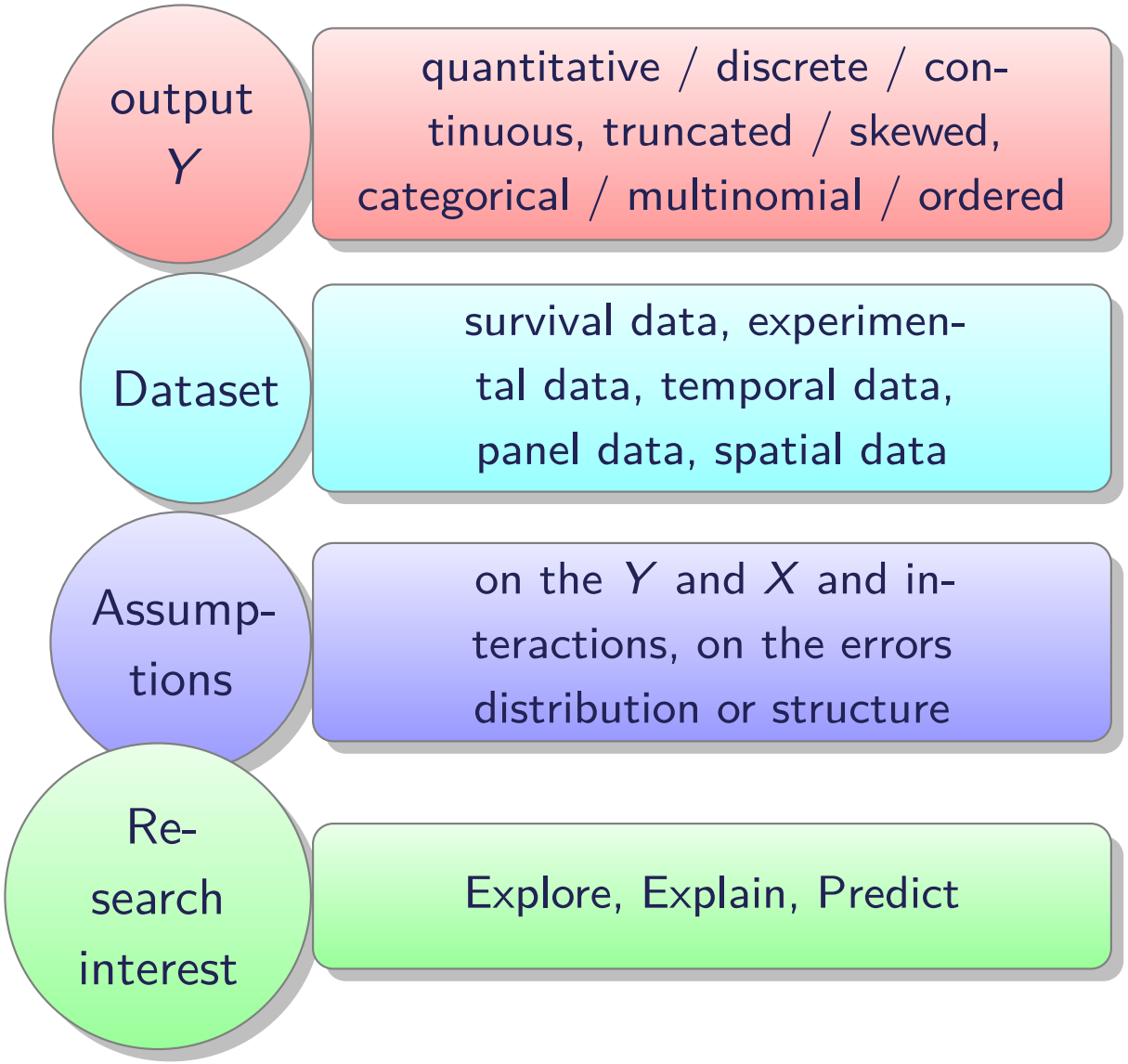


Figure 1: Linear Regression

How to choose the model?



Explanatory power

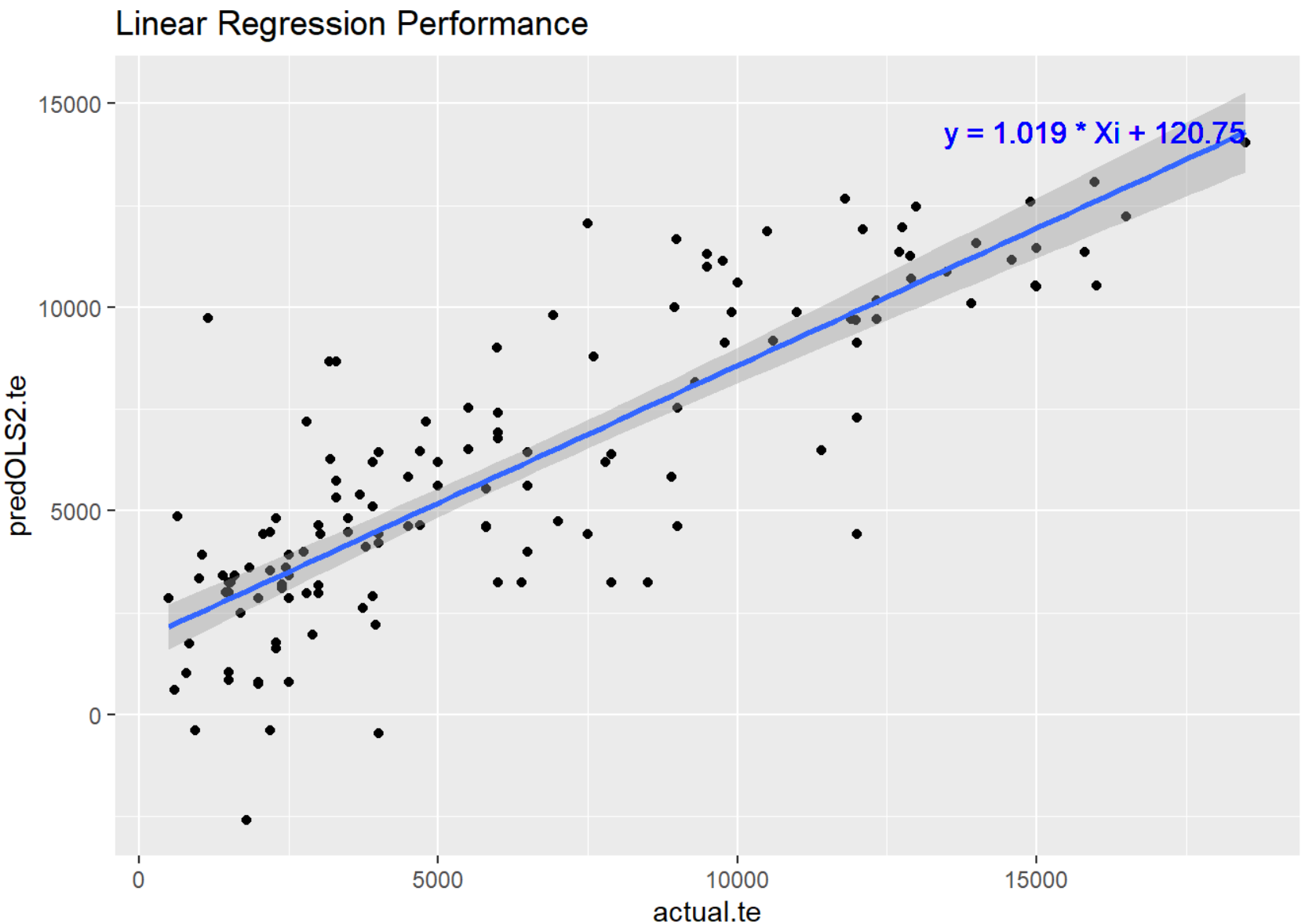
Theoritical Questions - Empirical answers

$$Y = \alpha + \beta \cdot X + \epsilon$$

Research Questions	ML Questions	Statistical Inference
1) What are the variables that are interrelated (or not)?	Is the parameter significantly different from zero?	$\Rightarrow$ *TEST*
2) What is the direction of the relationship?	Is the sign of the parameter positive or negative?	$\Rightarrow$ *TEST*
3) What is the magnitude involved (how much a change of one variable affects another)?	How to interpret the value of the parameter in terms of effect of the variable on the endogenous?	$\Rightarrow$ Prediction and TEST

Prediction power

Evaluating distance between prediction and unknown futur



Regression modeling

Getting Started

- Choosing the right kind of statistical analysis
  in order to explore / explain /predict
- Response Variable:
  - variable whose variation you are attempting to understand
  - y-axis of the graph
- Explanatory variables:
  - x-axis of the graph
  - which variation in the response variable is associated with variation in the explanatory variable.
- the measure:
  - **quantitative measurement** (real numbered value; discrete or continuous)
  - **categorical variable** (factor; ordered or un-ordered)

Regression statistical models:

The response variable

(a) Continuous:

- All explanatory variables *continuous* : **Multiple Linear Regression**
- All explanatory variables *categorical* : **Analysis of variance (ANOVA)**
- Explanatory variables both *continuous and categorical* : **Analysis of covariance (ANCOVA)**

(b) Proportion (or Binary class): **Logistic regression**

(c) Categorical (Multinomial): **Multinomial logistic analysis**

(d) Count: **Poisson models**

(e) Time at death: **Survival analysis**

(f) Panel data: **Panel regression**

Modelling regressions

Commun Objectives

- determine the values of the **parameters** in a specific model
- that lead to the **best fit of the model to the data**
- **minimal adequate model** to describe the data
- Model **is fitted to** data, not the reverse
- **Best model** produces the **least unexplained variation**,. . .
- . . . AND all the parameters should be **statistically significant**

Quick Test of Statistics and Linear regression model

Quick Test of Statistics and Linear regression model



Second Test on Linear regression model



Reminder of statistics

Reminder of statistics

See:

- Appendix in these slides
- Introduction to Statistical Learning with R (https://www.statlearning.com/)
- Elements of Statistical Learning, Hastie
- Introductory Statistics with R, Dalgaard

Regression Basics - Linear Regression

Linear Regression

Simple Regression

Regression analysis: using sample data to estimate **parameter values** and their **standard errors**

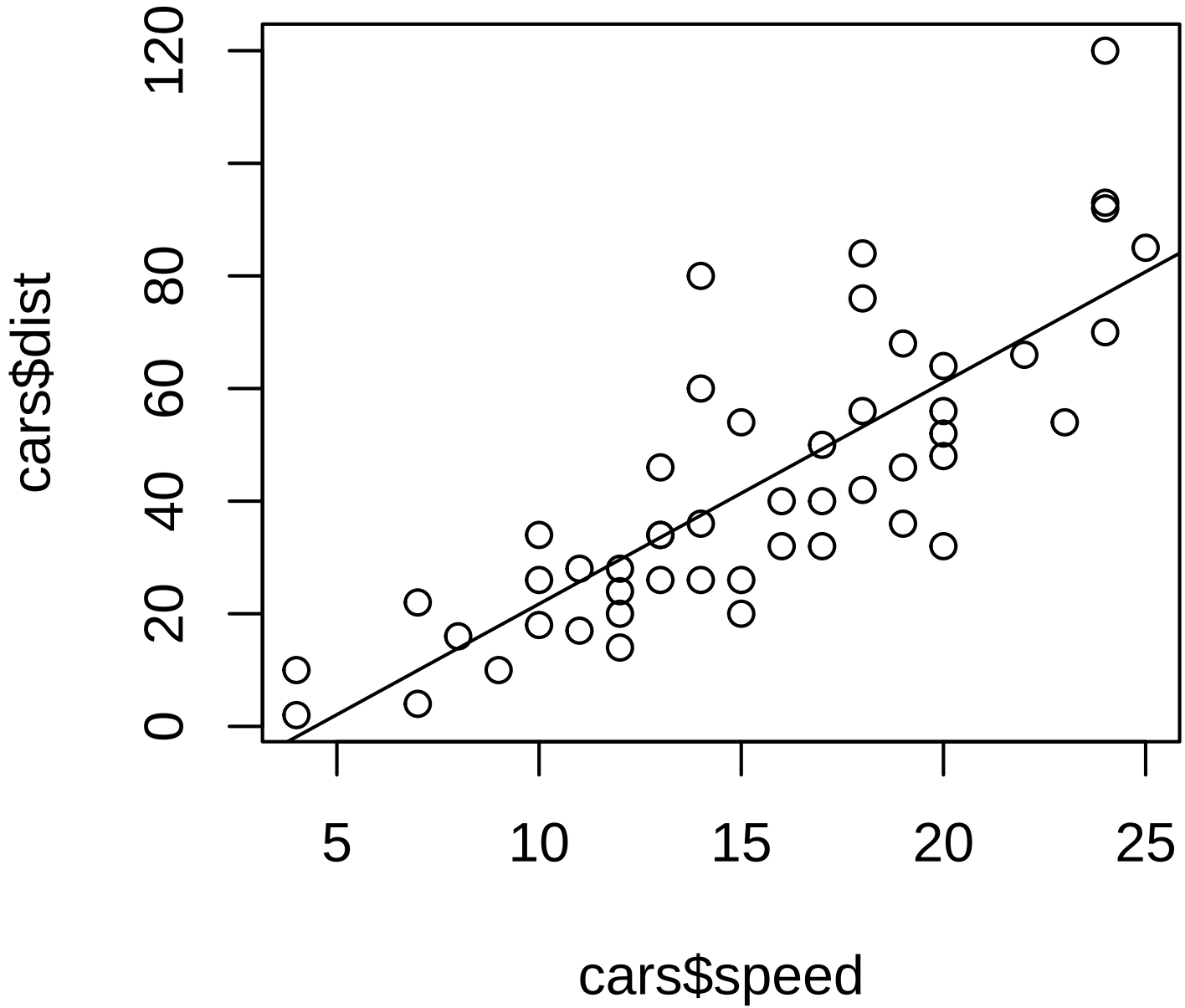
1) select a model which describes the relationship between the response variable and the explanatory variable(s)

Simple Regression simple linear model between one y and one x

$$y = \alpha + \beta x$$

- 2 parameters α and β
- the intercept α (the value of y when $x = 0$)
- the slope is β (the change in y divided by (and caused by) the change in x)
- causation is assumed by the model

```r
plot(cars$speed, cars$dist)
abline(lm(cars$dist~cars$speed))
```



Assumptions

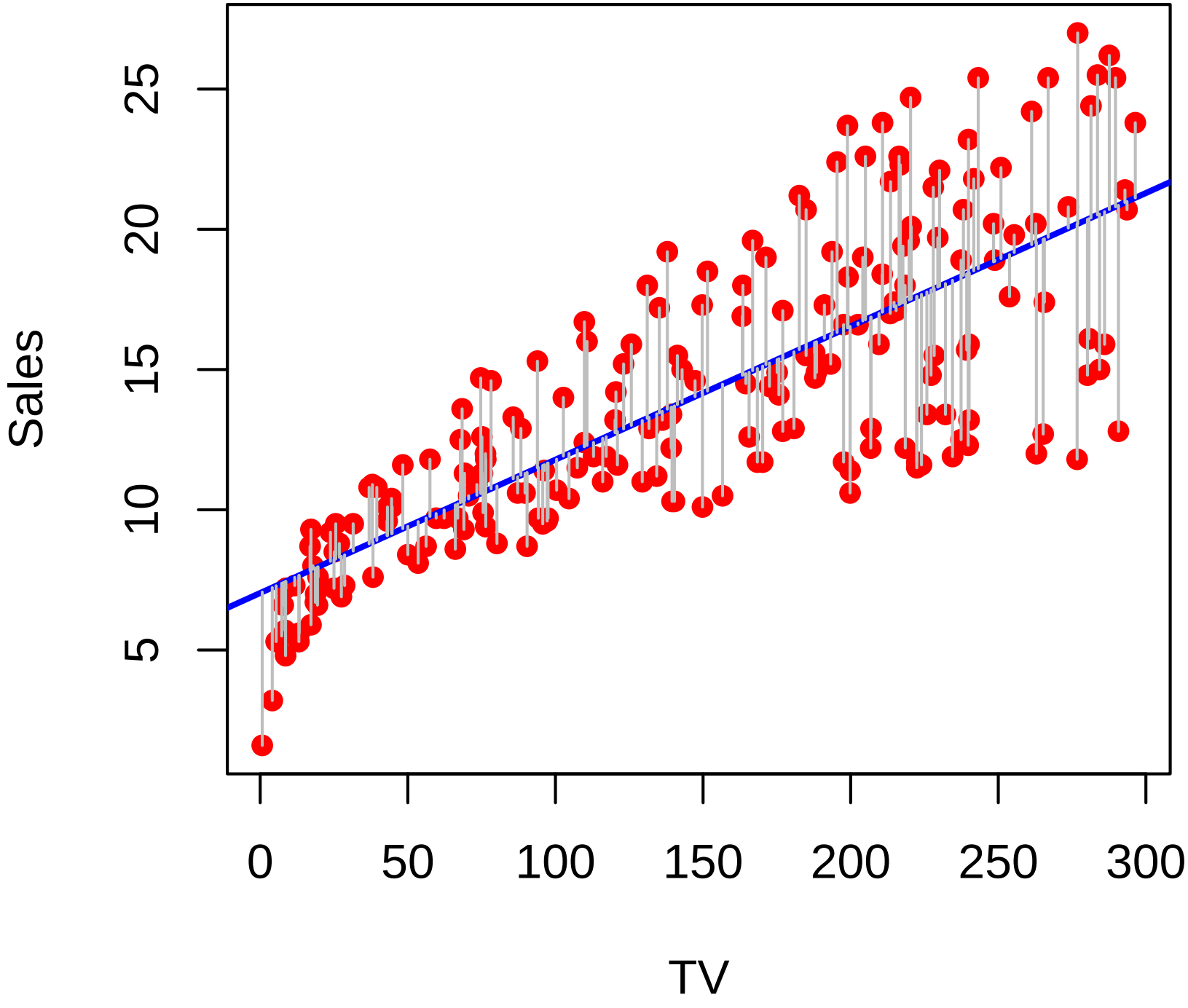
$$y = \alpha + \beta \cdot x + \epsilon$$

- The difference between a measured value of y and the value predicted by the model ($\hat{y}$ for the same value of x is called a residual (ϵ).
- The explanatory variable, x, is measured without error.
- The variance in residuals and y is constant (i.e. the variance does not change as y gets bigger).
- Residuals are measured on the scale of y (i.e. parallel to the y-axis).
- The residuals are normally distributed.

Estimation

Under these assumptions

- Maximum Likelihood and ordinary least squares methods give the same results
- Residuals: $e_i = y_i - \hat{y}_i = y_i - \alpha - \beta x$
- Measure of lack of fit is : $\sum e_i^2 = \sum (y_i - \alpha - \beta x)^2$
- Minimization of $\sum e_i^2$ produces efficient estimators of α and β (BLUE: best linear unbiased estimators)


Sales
5
10
15
20
25
0
50
100
150
200
250
300
TV

Analysis of variance in regression: SSY =SSR+SSE

- The best fit line pass through the mean point $(\bar{x}, \bar{y})$
- total variation in the response variable as $SSY = \sum(y_i - \bar{y})^2$
- residual variation or unexplained variation:
  - sum of squares of the residuals:
  - the error sum of squares, $SSE = \sum(y_i - \alpha - \beta x_i)^2$
- explained variation: $SSY - SSE$
- $R^2 = \frac{SSY - SSE}{SSY}$

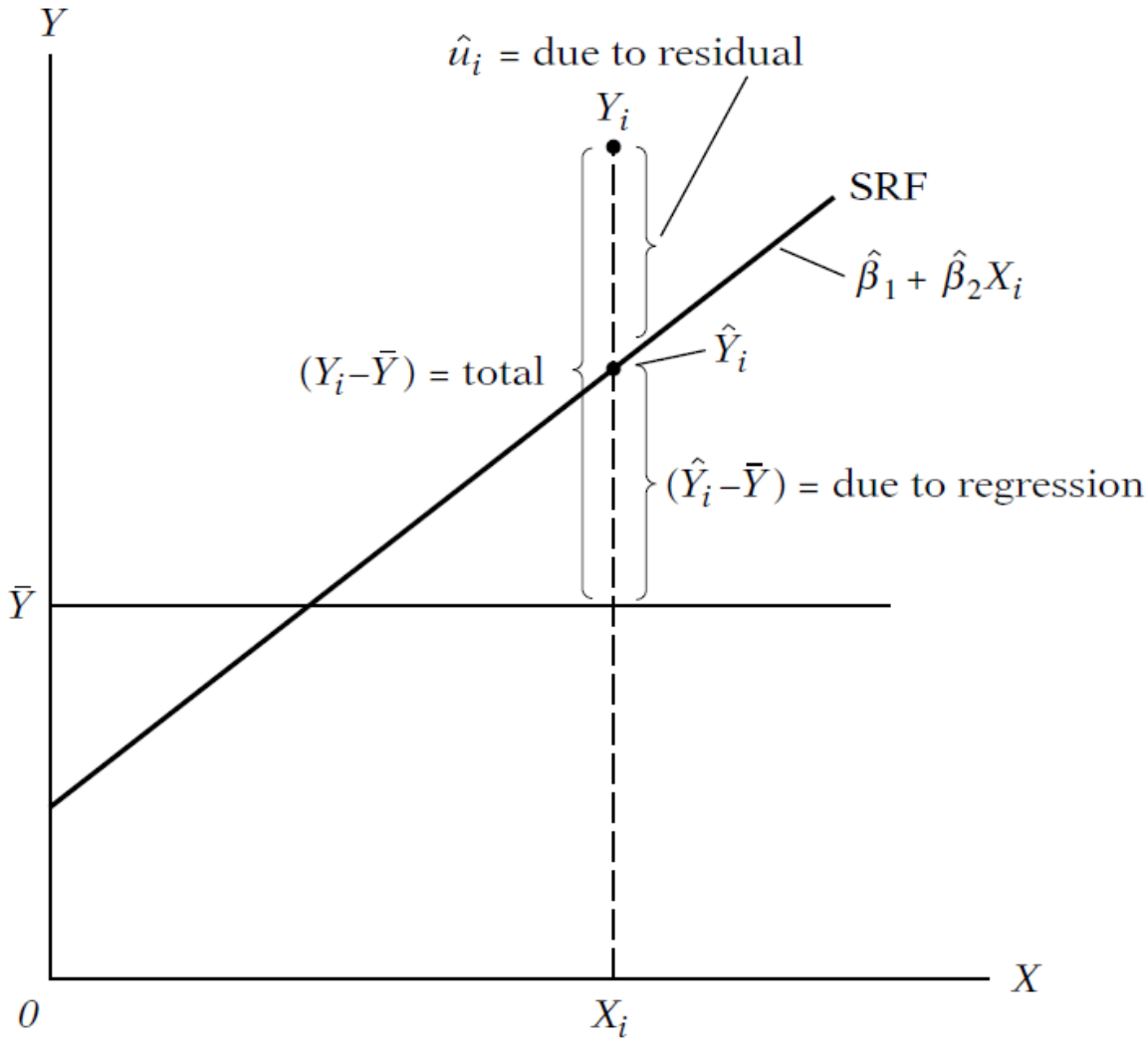


Figure 2: Variance decomposition

Unreliability estimates for the parameters

Unreliability associated with each of the estimated parameters:

- standard error of the intercept ($se(\hat{\alpha})$) and
- the standard error of the slope ($se(\hat{\beta})$)

The Principle of Parsimony (Occam's Razor)

The principle of parsimony (early 14th-century English nominalist philosopher, William of Occam)

> *given a set of equally good explanations for a given phenomenon, the correct explanation is the simplest explanation. ("Occam's razor")*

- in explaining something, assumptions must not be needlessly multiplied
- things not known to exist should not be postulated as existing

Guidelines for statistical modelling

For statistical modelling:

- as few parameters as possible
- linear preferred to non-linear models
- experiments relying on few assumptions preferred to those relying on many
- models should be pared down until they are minimal adequate;
- simple explanations should be preferred to complex explanations.

Seek simplicity, then distrust it.

- Einstein subtle modification to Occam's razor:
  'A model should be as simple as possible. But no simpler.'

- Oscar Wilde:
  'Truth is rarely pure, and never simple.'

Linear model with R

Model Formulae in R

The structure of the model:

- response variable ~ explanatory variable(s)
- ~: 'is modeled as a linear function of'

Exercise:

Define the model equation for the following formula

Model	Model formula
Null	`y~1`
Simple Regression	`y~x`
Regression through origin	`y ~ x-1`

- *gender* is a two-level categorical var.
- *genotype* is a four-level categorical variable

Model	Model formula
One-way ANOVA	`y ~ gender`
One-way ANOVA	`y ~ gender-1`
Two-way ANOVA	`y ~ gender + genotype`
Factorial ANOVA	`y~N * P * K`
Analysis of covariance	`y ~ x + gender`
Analysis of covariance	`y ~ x * gender`

Model	Model formula
Multiple regression	`y ~ x + z`
Multiple regression	`y~ x * z`
Multiple regression	`y ~ x + I(x^2) + z + I(z^2)`
Multiple regression	`y ~ poly(x,2) + z`
Multiple regression	`y~(x + z + w)^2`
Transformed response and explanatory variables	`log(y) ~ I(1/x) + sqrt(z)`

- Interactions between two two-level categorical variables:
  - number of interaction terms is $(a-1)(b-1)$
- Interactions between two continuous variables:
  - `x*z` means fit `x+z+x:z`
  - interaction term `x:z`: product vectors x and z
- Interactions between categorical and continuous variables:
  - separate slope and intercept for each level

Definition of a linear model

Linear models:

linear in terms of coefficients

$$y = a + bx$$

is coded `y~x`

$$y = a + bx - cx^2$$

is coded `y~x+I(x^2)`

non linear models:

$$y = exp(a + bx)$$

`lm()` function

- R has `lm()` for linear model

```
library(modelr)
sim1_mod <- lm(y ~ x, data = sim1)
coef(sim1_mod)
```

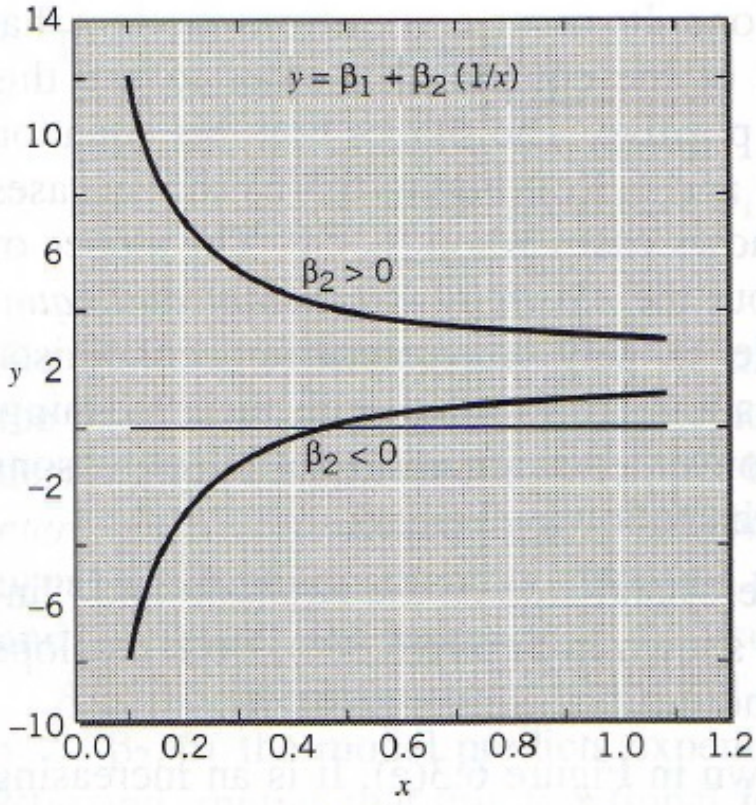
```
## (Intercept)           x
##    4.220822    2.051533
```

Functional Forms

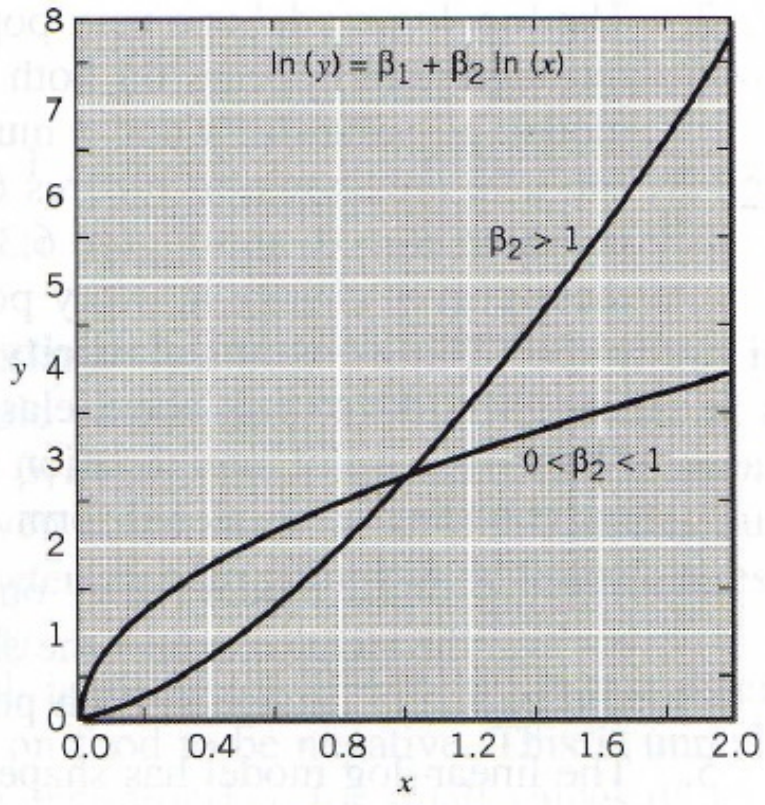
Type	Statistical Model	Slope	Elasticity
Linear	$y_t = \beta_1 + \beta_2 x_t + e_t$	β_2	$\beta_2 \frac{x_t}{y_t}$
Reciprocal	$y_t = \beta_1 + \beta_2 \frac{1}{x_t} + e_t$	$-\beta_2 \frac{1}{x_t^2}$	$-\beta_2 \frac{1}{x_t y_t}$
Log-Log	$ln(y_t) = \beta_1 + \beta_2 ln(x_t) + e_t$	$\beta 2 \frac{y_t}{x_t}$	β_2
Log-Linear (Exponential)	$ln(y_t) = \beta_1 + \beta_2 x_t + e_t$	$\beta_2 y_t$	$\beta_2 x_t$
Linear-Log (Semi-log)	$y_t = \beta_1 + \beta_2 ln(x_t) + e_t$	$\beta_2 \frac{1}{x_t}$	$\beta_2 \frac{1}{y_t}$
Log-Inverse	$ln(y_t) = \beta_1 + \beta_2 \frac{1}{x_t} + e_t$	$\beta_2 \frac{y_t}{x_t^2}$	$\beta_2 \frac{1}{x_t}$

Functional Forms

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Linear	$y_t = \beta_1 + \beta_2 x_t + e_t$	β_2	$\beta_2 \frac{x_t}{y_t}$
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Log-Log	$ln(y_t) = \beta_1 + \beta_2 ln(x_t) + e_t$	$\beta 2 \frac{y_t}{x_t}$	β_2
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Linear-Log (Semi-log)	$y_t = \beta_1 + \beta_2 ln(x_t) + e_t$	$\beta_2 \frac{1}{x_t}$	$\beta_2 \frac{1}{y_t}$
Log-Inverse	$ln(y_t) = \beta_1 + \beta_2 \frac{1}{x_t} + e_t$	$\beta_2 \frac{y_t}{x_t^2}$	$\beta_2 \frac{1}{x_t}$

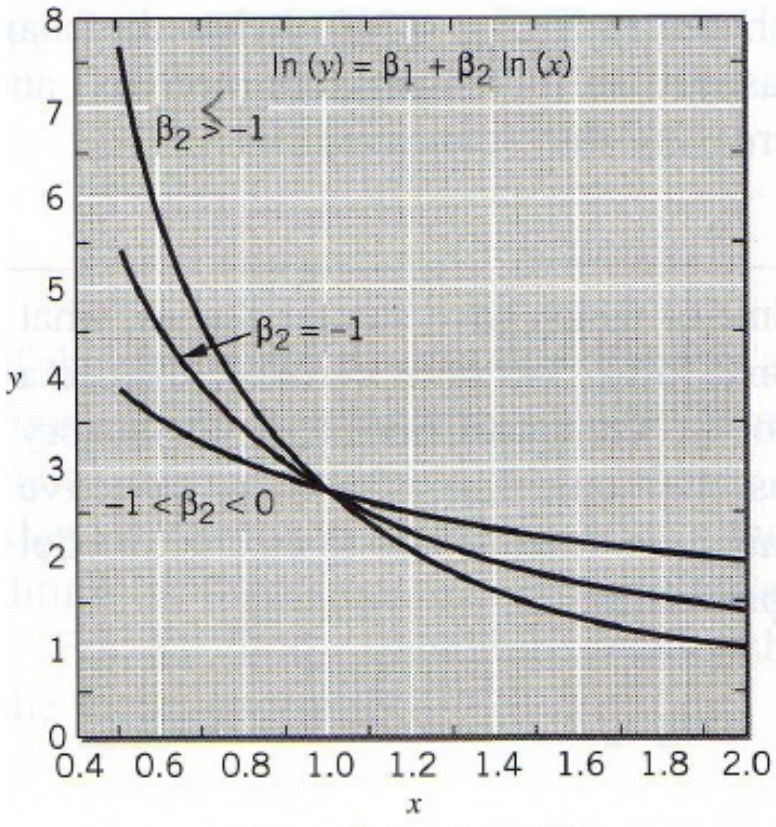


(*a*) Type 2: Reciprocal Model

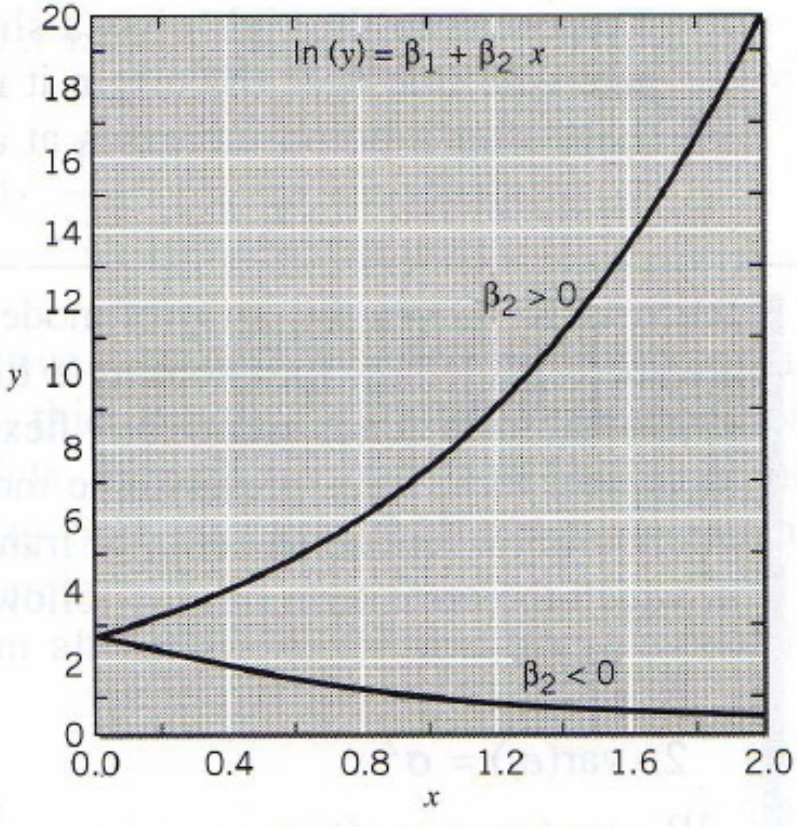


(*b*) Type 3: Log–Log Model

Figure 3: Econometric modelling

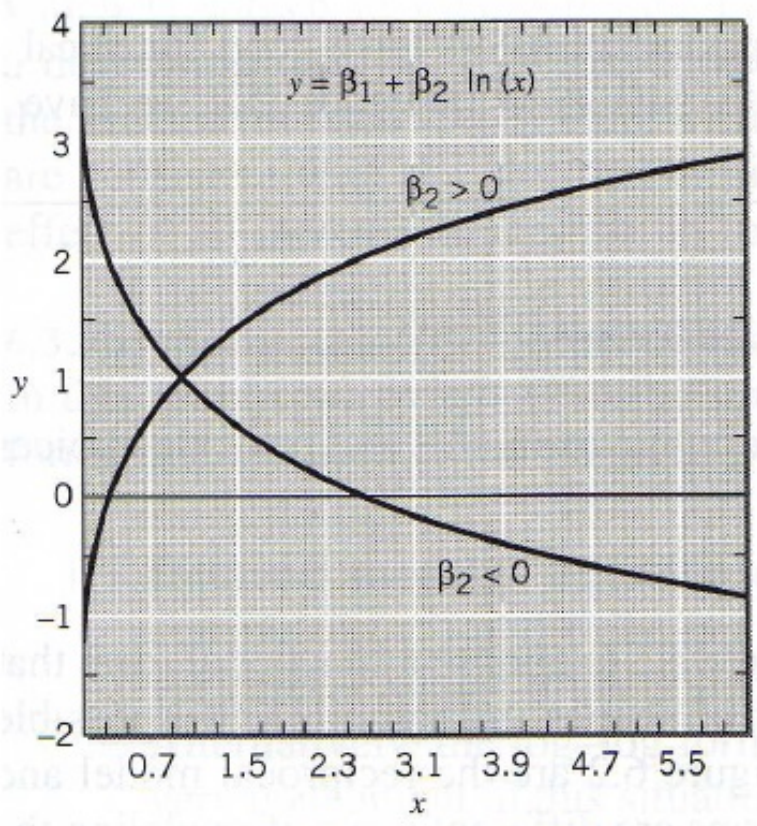


(c) Type 3: Log–Log Model

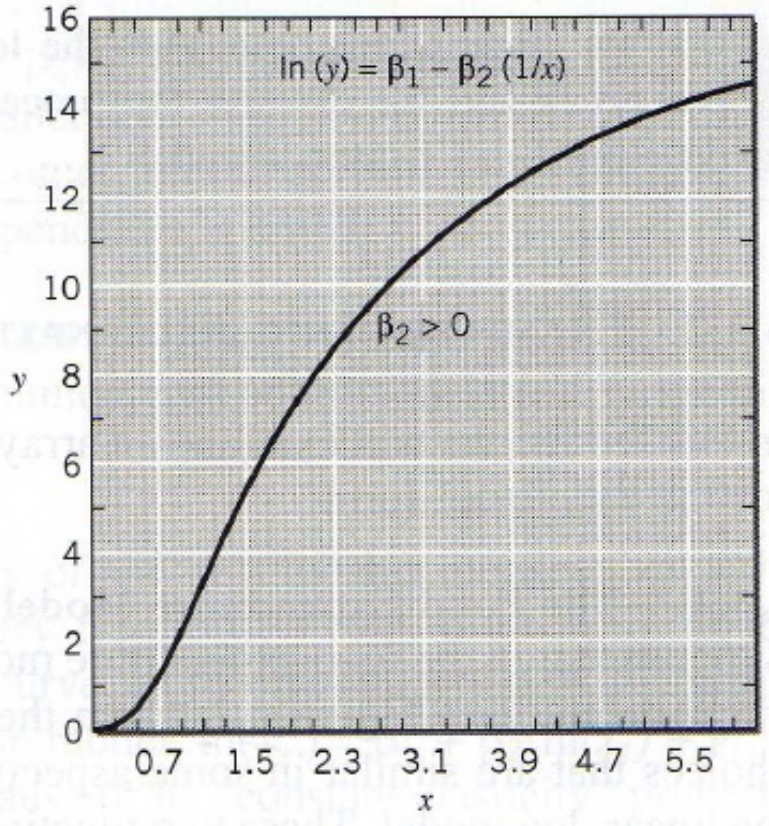


(d) Type 4: Log–Linear Model

Figure 4: Econometric modelling



(*e*) Type 5: Linear–Log Model



(*f*) Type 6: Log–Inverse Model

Figure 5: Econometric modelling

Diving into simple linear model

Simple linear model with Categorical X Variables

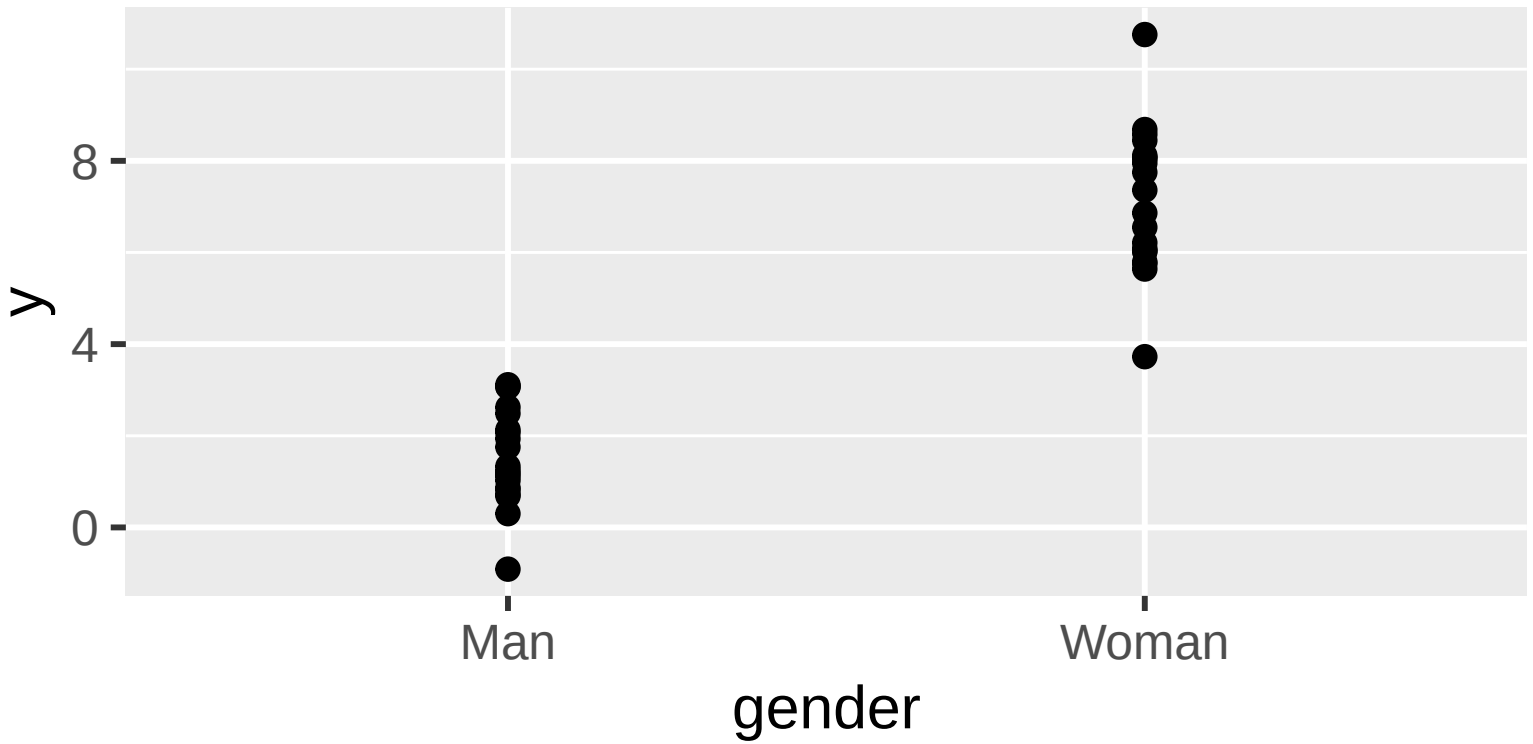
- Using `Gender` has no meaning:

$$y = \alpha + \beta \times \texttt{Gender}$$

  `Gender` has no numeric values

- Instead convert `Gender` into **Boolean** `Gender.Woman` which takes value 1 for women and zero otherwise
- why we do **not create** a `Gender.man` column?
  It would create a column that is perfectly predictable
  (i.e., `Gender.man` = 1 - `Gender.Woman`)
- Regular way with R to deal categorical variables:
  `factor(Gender)`

Example



Fitted linear model

$$y = \alpha + \beta \times \texttt{Gender.Woman}$$

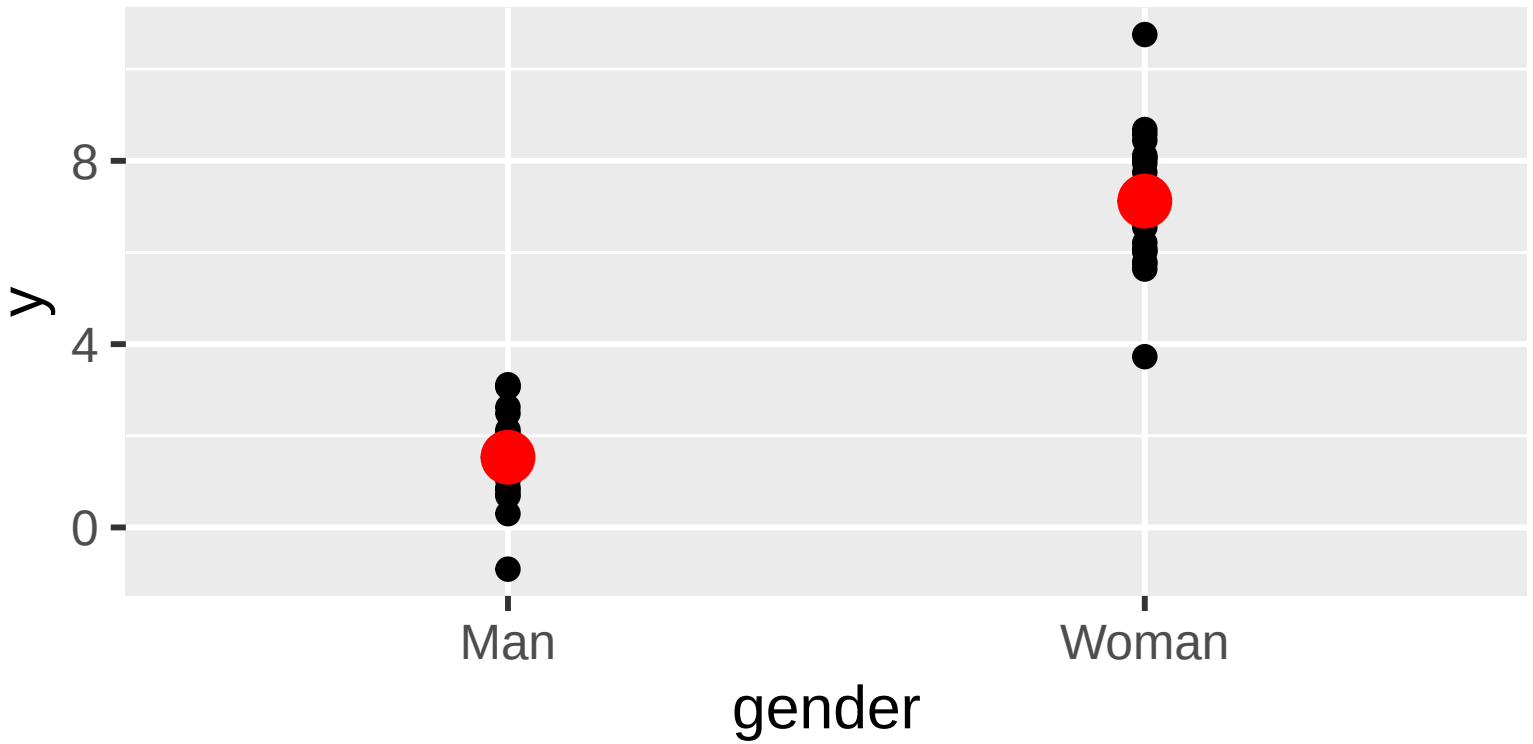
```
mod2 <- lm(y ~ gender, data = sim2)
mod2
```

```
##
## Call:
## lm(formula = y ~ gender, data = sim2)
##
## Coefficients:
## (Intercept)  genderWoman
##       1.532        5.590
```

Prediction with categorical variable

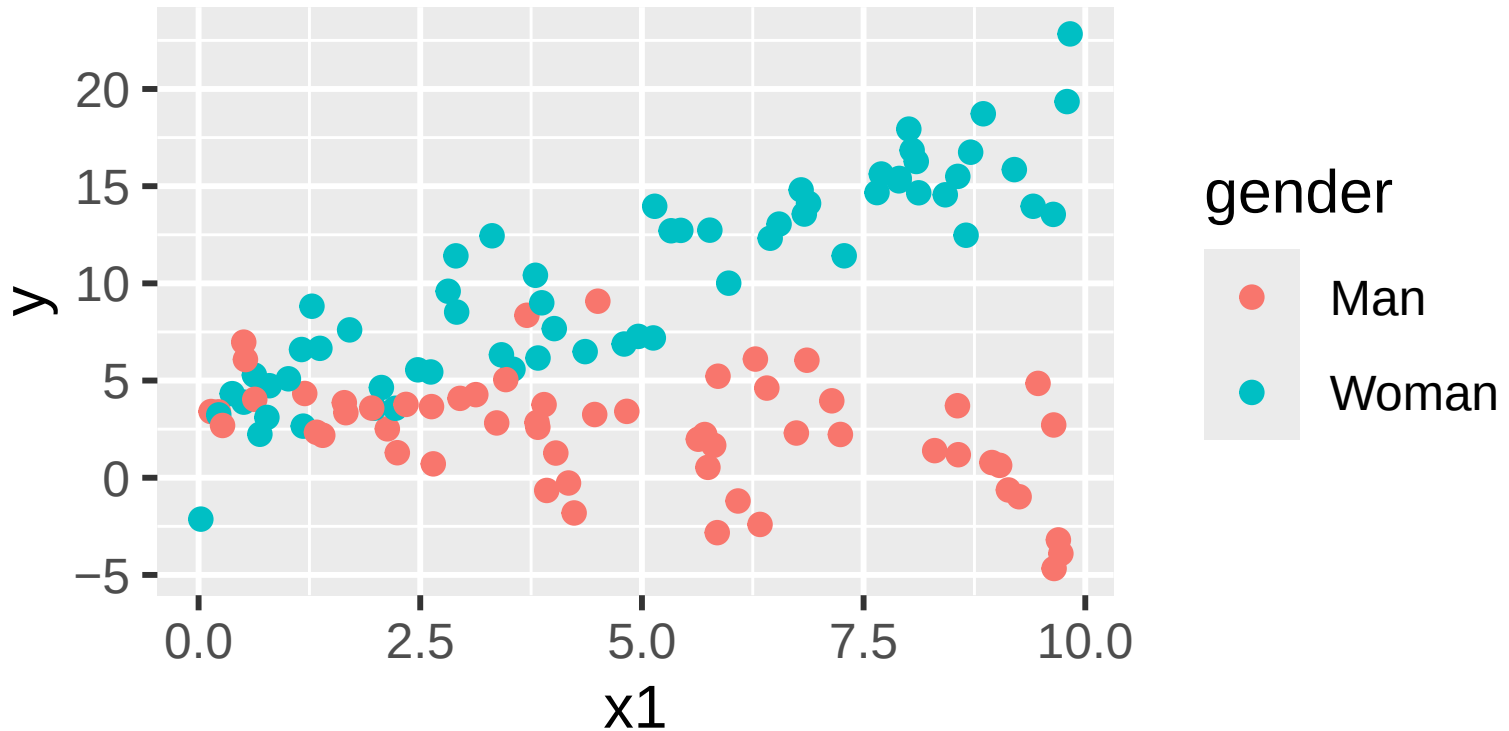
- a model with a categorical X will predict the mean value for each category
- Why? Because the **mean minimizes the root-mean-squared distance**

Prediction with categorical variable



Interactions (Continuous and Categorical)

- What happens when you combine a continuous and a categorical variable?



Interaction in the model

Model equation

- without interaction:

  $y = \alpha + \beta_1 x_1 + \beta_2 \texttt{Gender.Woman}$

  which is:

  if $\texttt{Gender.Woman} = 0$: $y = \alpha + \beta_1 x_1$

  if $\texttt{Gender.Woman} = 1$: $y = \alpha + \beta_1 x_1 + \beta_2$

- with interaction:

  $y = \alpha + \beta_1 x_1 + \beta_2 \texttt{Gender.Woman} + \beta_{1,2} x_1 \texttt{Gender.Woman}$

  which is:

  if $\texttt{Gender.Woman} = 0$: $y = \alpha + \beta_1 x_1$

  if $\texttt{Gender.Woman} = 1$: $y = \alpha + \beta_1 x_1 + \beta_2 + \beta_{1,2} x_1$

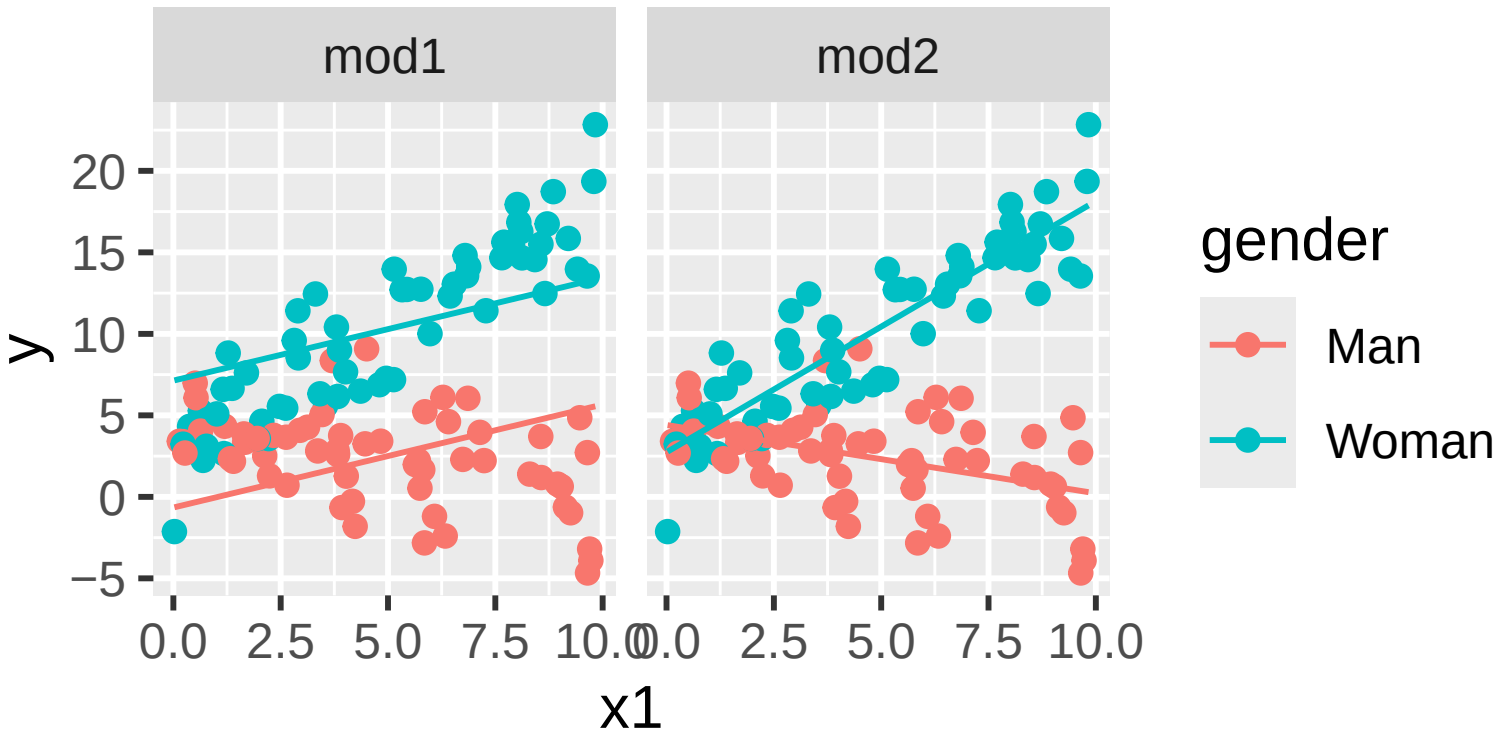
  $\Leftrightarrow y = \alpha + \beta_2 + (\beta_1 + \beta_{1,2}) x_1$

Interaction in the model

The formula argument of `lm()` specifies with (*) or without (+) interaction models:

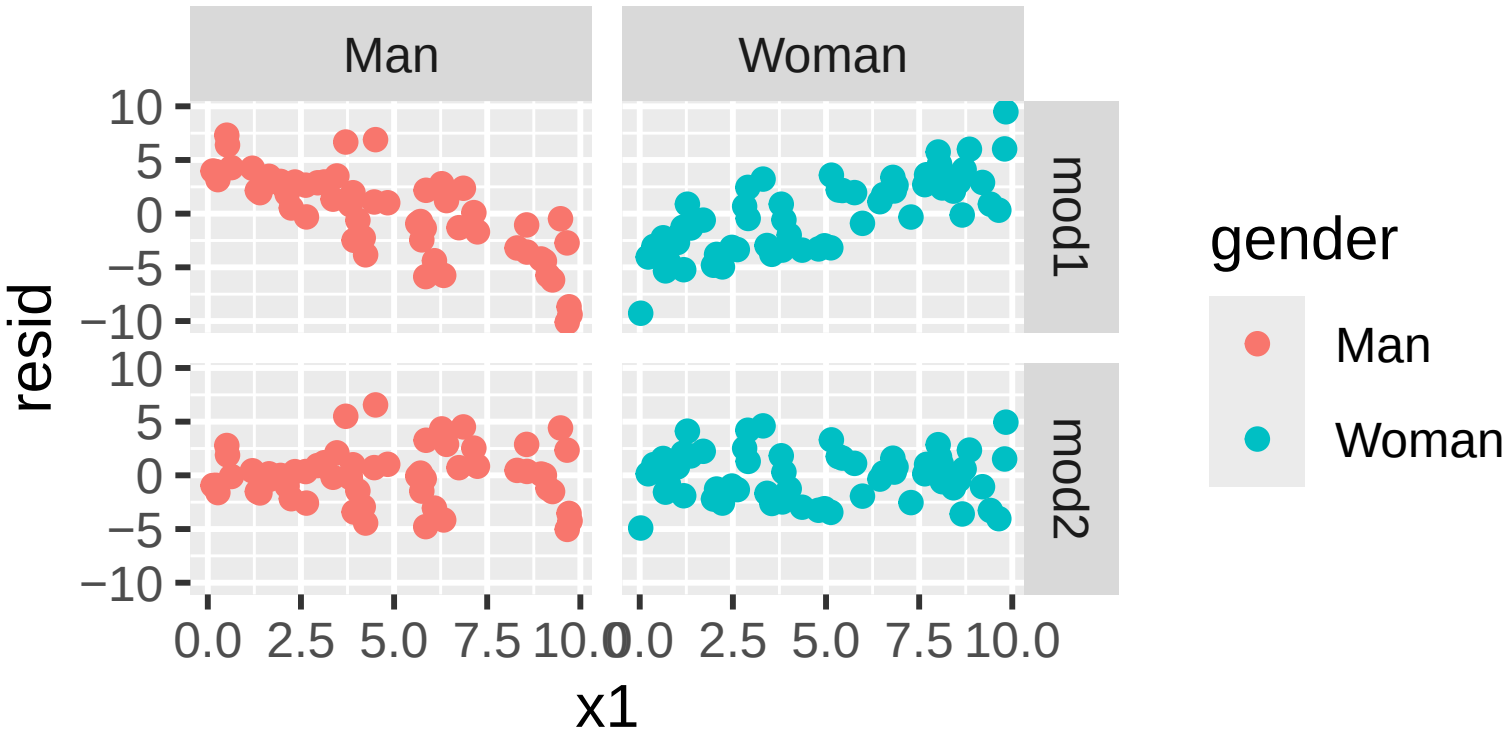
```
# Without interaction : +
mod1 <- lm(y ~ x1 + gender, data = sim3)
# With interaction *
mod2 <- lm(y ~ x1 * gender, data = sim3)
```

Vizualisation of interaction



Which model is better for this data?

- We can take look at the residuals



Conclusion

- Is there's a precise way to tell which of mod1 or mod2 is better?

  Yes using more math and stat background
- **Lets do that in an exercise**

Appendix

Solution to Exercise

Solution to Exercise:

Model	Model formula	
Null	`y~1`	1 is the intercept it is the overall mean y
Regression	`y~x`	x is a continuous expl. var.
Regression through origin	`y ~ x-1`	Do not fit an intercept

| Model | Model formula | |
|:———:|:———————-:|:———————————-| | One-way ANOVA | `y ~ gender` | *gender* is a two-level categorical var. | | One-way ANOVA | `y ~ gender-1` | as above, but no intercept | | | | (two means rather than mean and difference) | | Two-way ANOVA| `y ~ gender + genotype` | *genotype* is a four-level categorical variable| | Factorial ANOVA | `y~N * P * K` | N, P and K are two-level factors to be fitted along with all their interactions | |Analysis of covariance| `y ~ x + gender` | A common slope for y against x but with two intercepts, one for each *gender* | |Analysis of covariance| `y ~ x * gender` | Two slopes and two intercepts |

Model	Model formula	
Multiple regression	`y~x + z`	Two continuous explanatory variables, flat surface fit
Multiple regression	`y~ x * z`	Fit an interaction term as well ($x + z + x : z$)
Multiple regression	`y ~ x + I(x^2) + z + I(z^2)`	Fit a quadratic term for both x and z
Multiple regression	`y ~ poly(x,2) + z`	Fit a quadratic polynomial for x and linear z
Multiple regression	`y~(x + z + w)^2`	Fit three variables plus all their interactions up to two-way
Transformed response and explanatory variables	`log(y) ~ I(1/x) + sqrt(z)`	All three variables are transformed in the model

Statistics reminder

What is a p-value?

What is a p-value?

- PV: risk rejecting the null assumption
- if the risk (PV) is small: **we reject the null assumption**

Example of H_0

- null assumption: H_0:
  $H_0 : \mu_X = 10$ vs. $H_1 : \mu_X \neq 10$
  $H_0 : \mu_X > 10$ vs. $H_1 : \mu_X \leq 10$
  $H_0 : \mu_X < 10$ vs. $H_1 : \mu_X \geq 10$
- null assumption: H_0:
  $H_0 : \mu_X = \mu_Y$ vs. $H_1 : \mu_X \neq \mu_Y$
  $H_0 : X \sim \mathcal{N}(\mu_X, \sigma_X)$ vs. $H_1 : X \neq \mathcal{N}(\mu_X, \sigma_X)$
  ...

t-test or Student test

```
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00
```

Can we infer based on this sample that for the population (the cars on this road) : the mean of speed lower than 10 mph?

```r
#par(mfrow=c(2,2))
plot(cars$speed)
boxplot(cars$speed)
hist(cars$speed, main = "")
```

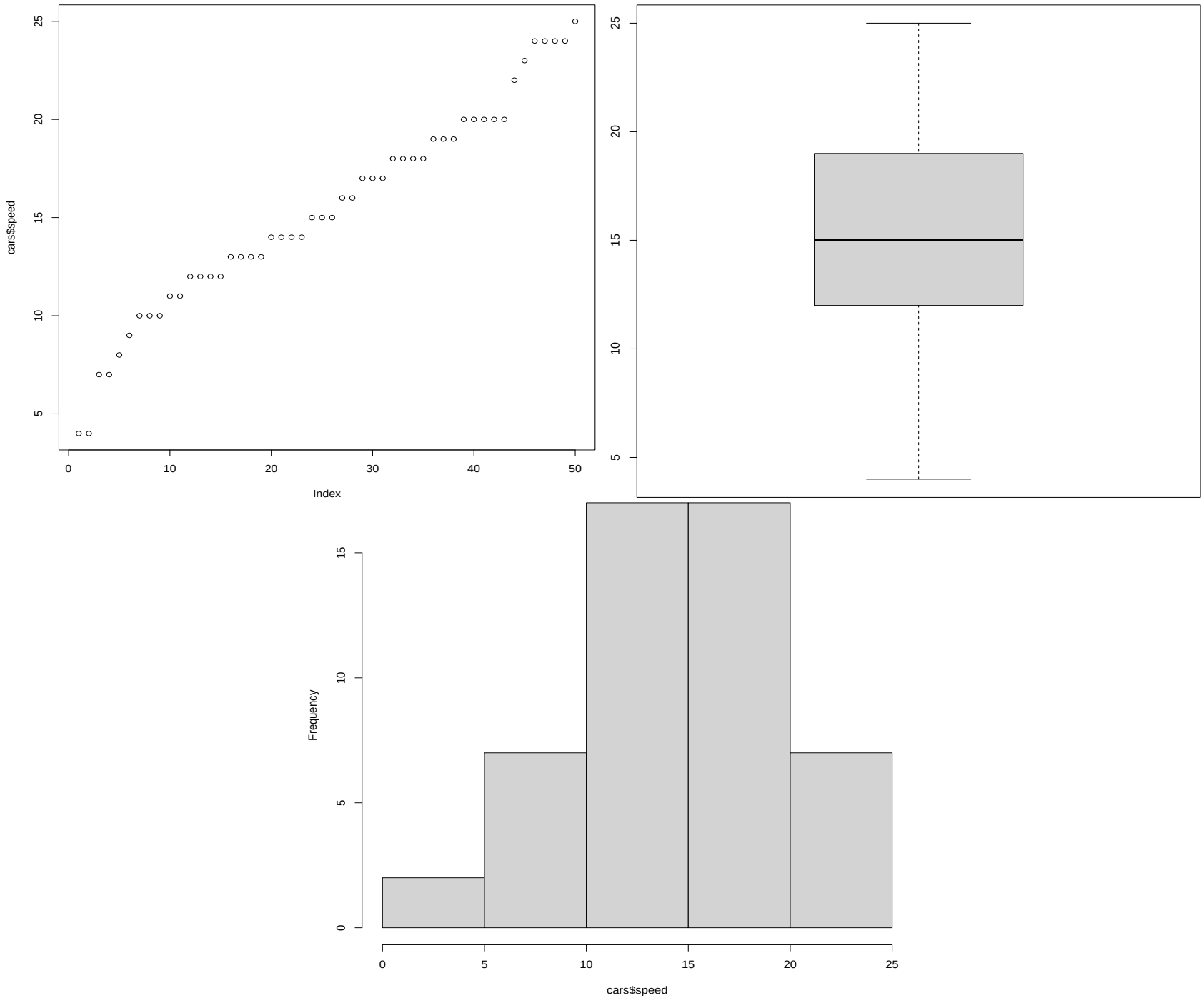


Figure 6: Speed figures

T-test

$$H_0 : \mu_{H_0} \leq 10$$

vs

$$H_1 : \mu_{H_1} > 10$$

Under usual statistical assumptions, bc $n = 50$

$$\frac{\bar{X} - \mu_{H_0}}{\hat{\sigma}/\sqrt{n}} \sim \mathcal{S}t(n-1)$$

t-test with R

```
mu_0 <- 10

t.test(cars$speed, mu=mu_0, alternative = 'greater')
```

```
##
##  One Sample t-test
##
## data:  cars$speed
## t = 7.2213, df = 49, p-value = 1.507e-09
## alternative hypothesis: true mean is greater than 10
## 95 percent confidence interval:
##  14.1463     Inf
## sample estimates:
## mean of x
##      15.4
```

The P-value

A **p-value** is an estimate of:

the probability that a particular observation (or a more extreme) could have occurred by chance, if the null hypothesis were true

- PV measures the "credibility" of the null hypothesis

If something is sufficiently unlikely to have occurred by chance (say, $p<0.05$), we say that it is statistically significant.

For example, in comparing two sample means, where the null hypothesis is that the means are the same ($H_0 : \mu_1 = \mu_2 \quad vs \quad H_1 : \mu_1 \neq \mu_2$):

- **low p-value** (pv $< 5\%$) means that the hypothesis is unlikely to be true and the **difference is statistically significant**.
- large p-value (say $p = 0.23$) means that there is no compelling evidence on which to reject the null hypothesis
- Of course, saying 'we do not reject the null hypothesis' and 'the null hypothesis is true' are two quite different things

Two samples

The classical tests for two samples include:

- comparing two variances (Fisher's F test, `var.test`)
- comparing two sample means with normality assumption (Student's t test, `t.test`)
- comparing two means with non-normality assumption (Wilcoxon's rank test, `wilcox.test`)
- comparing two proportions (the binomial test, `prop.test`)
- correlating two variables (Pearson's or Spearman's rank correlation, `cor.test`)
- testing for independence of two variables in a contingency table (chi-squared, `chisq.test`, or Fisher's exact test, `fisher.test`).

Correlation and covariance

With two continuous variables, x and y:

$$cov(x, y) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}))$$

$$r = \frac{cov(x, y)}{\sqrt{s_x^2 s_y^2}}$$

when x and y are independent

- then they are not correlated
- then the covariance between x and y is 0
- $E(xy) = E(x) \times E(y)$
- $r \in [-1, 1]$

Types of Statistical Model

Statistical modelling involves the selection of a minimal adequate model from a potentially large set of more complex models, using stepwise model simplification.

Model	Interpretation
Saturated model	One parameter for every data point *Fit*: perfect *Degrees of freedom*: none *Explanatory power*: none
Null model	Just one parameter, the overall mean $\bar{y}$ *Fit*: none; $SSE = SSY$ *Degrees of freedom*: $n - 1$ *Explanatory power*: none

Model	Interpretation
Maximal model	Contains all p factors, interactions and covariates of any interest. Many of the model's terms are likely to be insignificant *Degrees of freedom*: $n - p - 1$ *Explanatory power*: it depends
Minimal adequate model	Simplified model with $0 \leq p' \leq p$ param. *Fit*: $<$ max. model, but not significantly so *Degrees of freedom*: $n - p - 1$ *Explanatory power*: coef. of determination R^2 or Likelihood Ratio

The stepwise progression:

- from the saturated model (or the maximal model, whichever is appropriate) through a series of simplifications to the minimal adequate model
- stepwise is combination of forward and backward progression

- Models are representations of reality that should be both accurate and convenient.
- Trade-off between model's realism, generality and holism and parsimony

Preferences and Parsimony (other things being equal), we prefer:

- model with $n - 1$ parameters to n parameters
- model with $k - 1$ explanatory variables to k
- linear model to curved model
- model without a hump to a model with a hump
- model without interactions to model with interactions

References

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